Group 4 Project 1: Interim Report

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Introduction

This report analyzes individual-level data for Oregon residents from the 2020–2024 American Community Survey (ACS) 5-year Public Use Microdata Sample (PUMS) (1). The ACS PUMS provides a representative subsample of the U.S. population with detailed demographic, socioeconomic, and labor market information, enabling analysis at the individual level while preserving confidentiality. All analyses account for the survey design through the use of person weights, and income measures are adjusted to constant dollars using the Census Bureau's adjustment factor.

Following the OSEMN (Obtain, Scrub, Explore, Model, and Interpret) framework (2), we conduct exploratory analyses and develop statistical models to examine relationships between key variables of interest. The primary focus of this report is to identify factors associated with income and other socioeconomic outcomes in Oregon. Specifically, we investigate (1) predictors of income using penalized linear regression, (2) individual characteristics associated with lacking health insurance coverage, and (3) the relationship between income and transportation-related factors such as commute time, vehicle access, and location. These analyses aim to provide insight into the demographic and structural factors that contribute to economic outcomes within the state.

Question 1: Can income be predicted from selected census variables?

Exploratory Data Analysis

We begin by exploring the relationship between income and a few key predictors: age, education, gender, race, place of birth and number of household members.

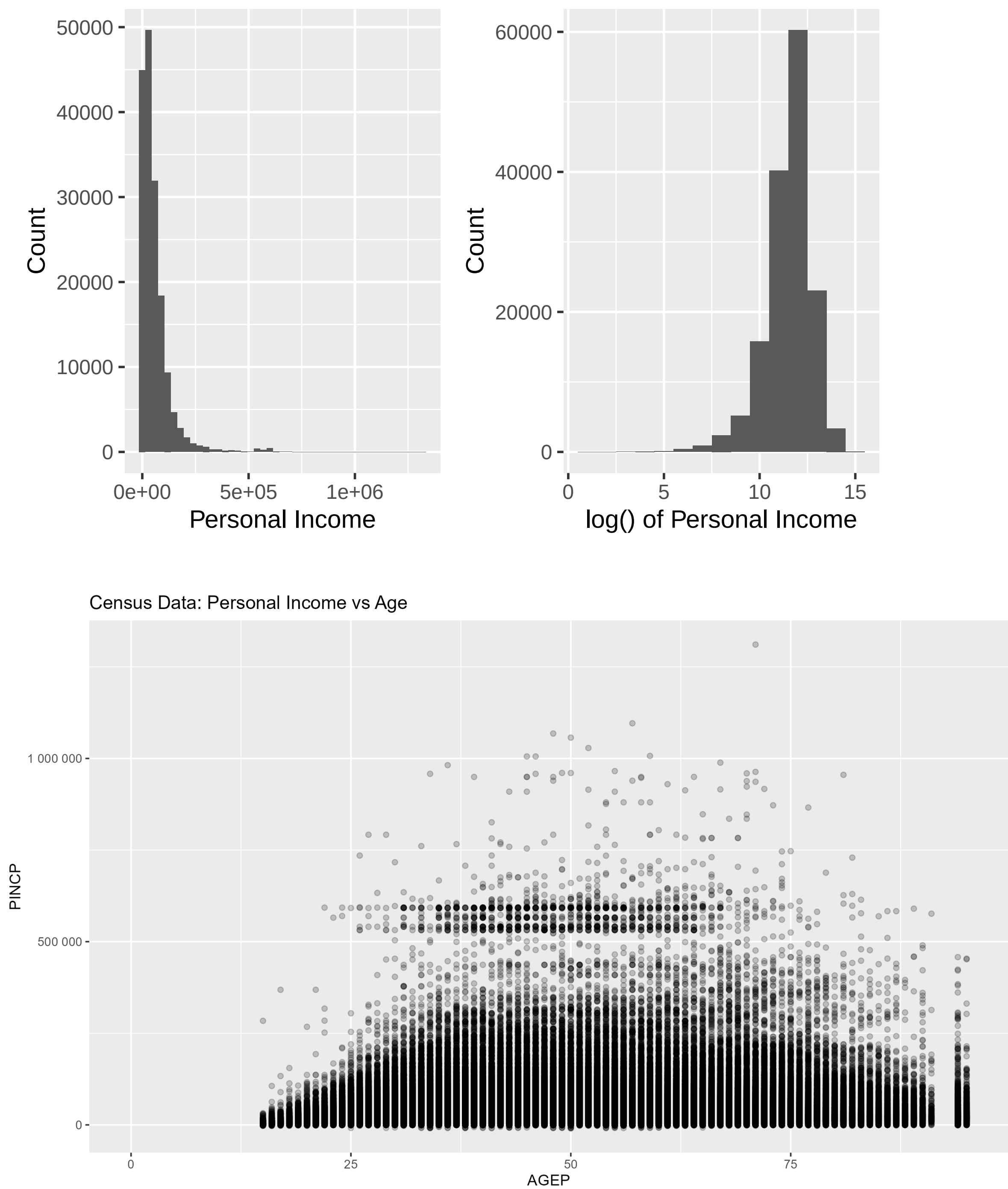


Figure 1: Distribution of personal income (PINCP) and log-transformed income, along with income versus age.

The above summary statistics indicate that income (PINCP) is highly right-skewed, with a small number of individuals earning substantially more than the median. Because of this skewness, we applied a log transformation to the income response when evaluating linear model fit for the final report. The scatterplot of income versus age suggests a nonlinear relationship: income tends to increase with age early in life, plateau in middle age, and decline slightly at older ages. This pattern suggests the inclusion of a quadratic age term in the regression model.

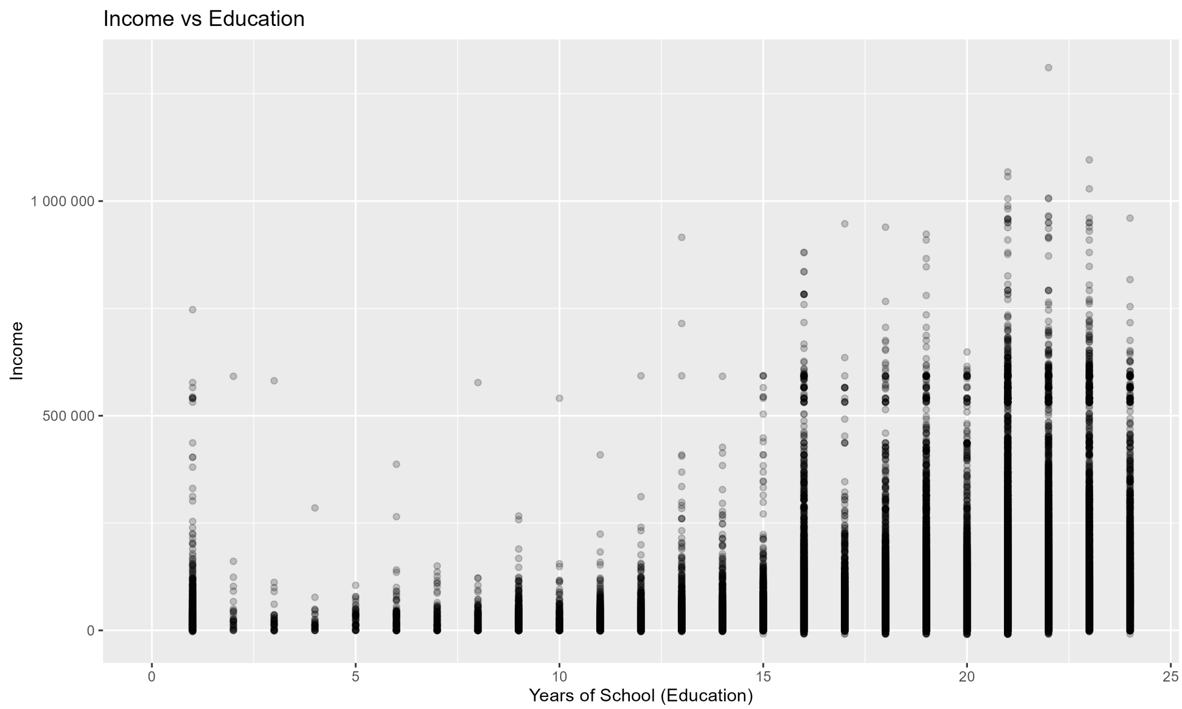
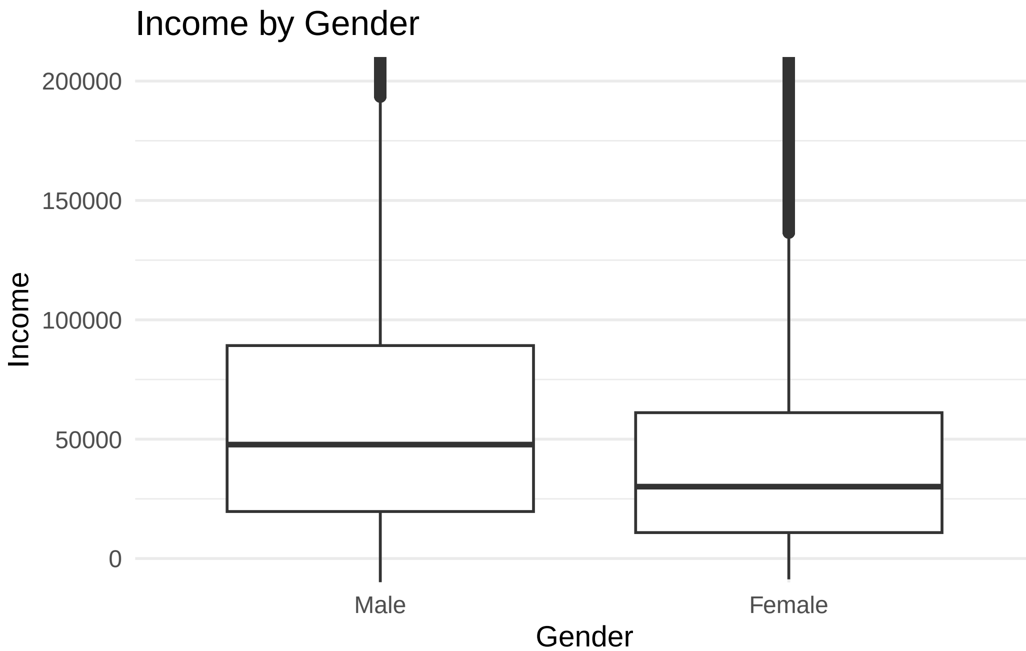
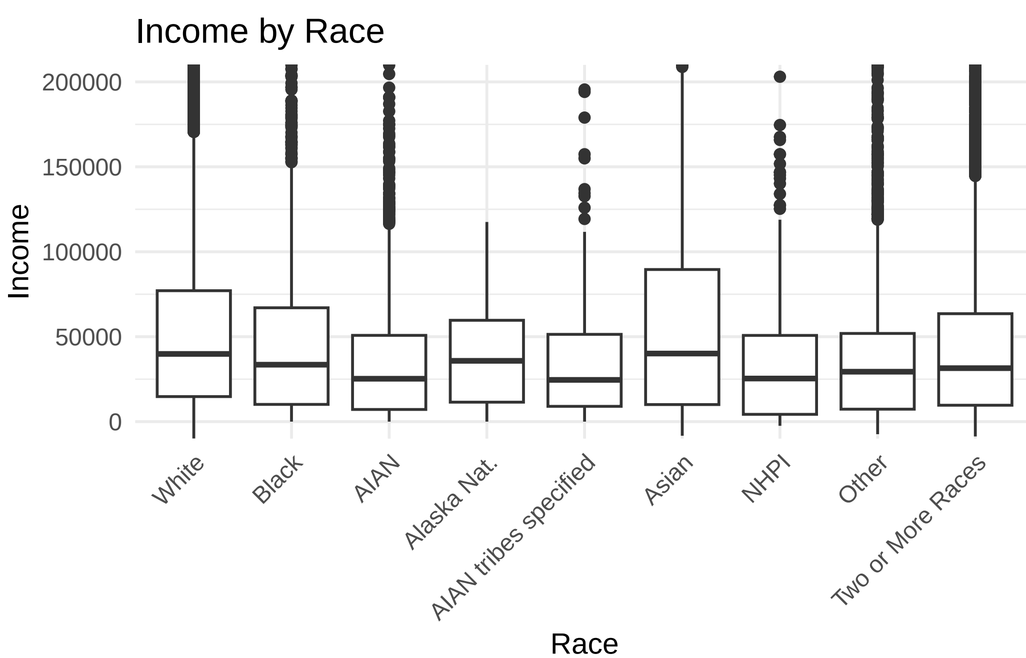


Figure 2: elationship between income and years of education.





attainment corresponding to higher earnings. Boxplots of income by gender and race reveal differences in median income across groups, supporting their inclusion as categorical predictors. Overall, the exploratory analysis suggests that income is influenced by multiple demographic and socioeconomic factors, and that nonlinear effects, particularly for age, should be incorporated into the model.

Q1: Model Selection

Based on our analysis of the data, we fit multiple linear regression models to assess how well income can be predicted from the selected variables. The initial full model is:

$$Income = Age + Age^2 + Education + Gender + Race + Birthplace + Householdmembers$$

We also included models of the log income as well. Income is modeled as a continuous response, and predictors include both continuous variables (age, education, household members) and categorical variables (gender, race, birthplace). A quadratic term for age is included to capture the nonlinear relationship observed in the exploratory analysis.

A reduced model excluding birthplace and household members is also considered due to weak initial evidence of its importance:

$$Income = Age + Age^2 + Education + Gender + Race$$

Q1: Model Checking and Selection

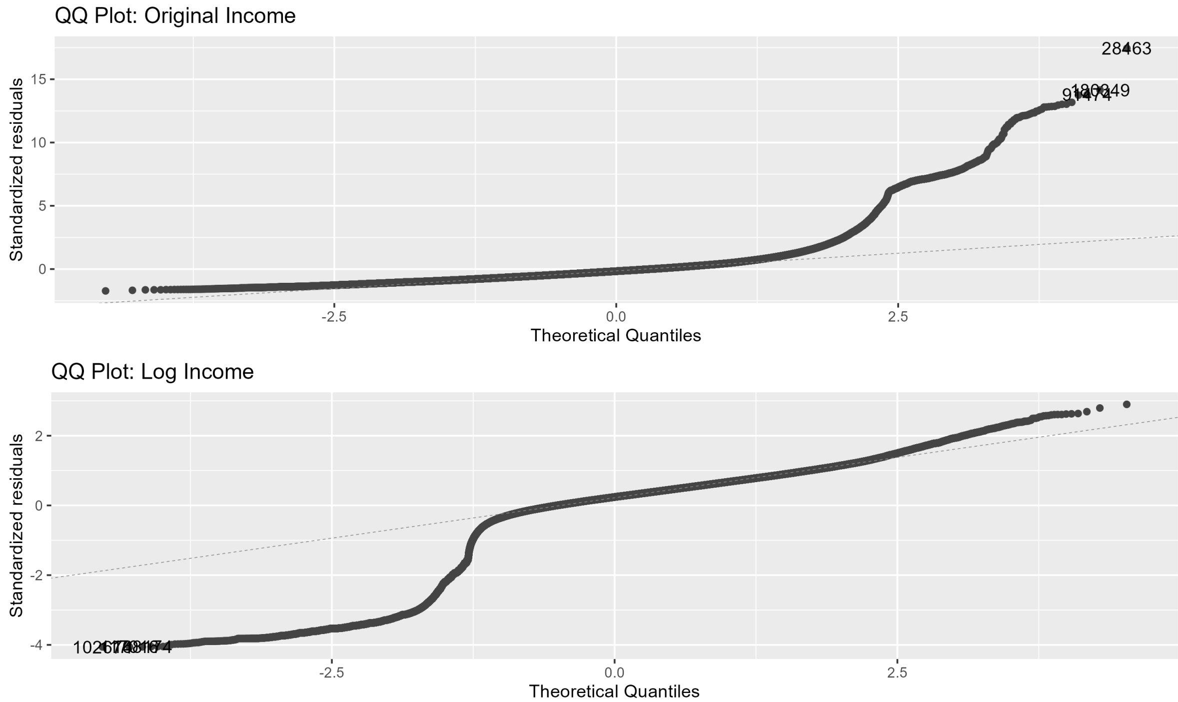


Figure 3: QQ plots of residuals from linear models using original and log-transformed income.

Model diagnostics were conducted to assess the validity of the linear regression assumptions. Residual plots indicated non-constant variance and right-skewness, which is consistent with the skewed distribution of income. We also conducted and plotted a log transformation of income. The residual plots suggest improves overall fit, although the untransformed model is retained for interpretability. Multicollinearity was assessed using variance inflation factors (VIF). All predictors showed low VIF values (near 1), except for age and age squared, which exhibited moderate collinearity. This is expected due to the polynomial relationship and does not invalidate the model. Centering age can reduce this collinearity but does not change model fit.

Alongside the linear model approach, we also evaluated a larger model using penalized linear regression. We separated Personal Income (PINCP) and all other variables in the dataset, leaving out other income related variables, then ran a series of lasso methods on the data to find a model of best fit and of a reduced model (yet still expanded from our linear approach) for better performance.

While the initial linear regression models were based on a limited set of predictors identified through exploratory analysis, they may omit relevant variables and fail to capture the full complexity of the data. Given the large number of available predictors in the ACS dataset, we expanded the modeling approach to consider a broader set of variables. To manage the resulting increase in dimensionality and potential multicollinearity, we employed penalized linear regression using the LASSO method. LASSO allows us to identify a more targeted set of predictors while maintaining predictive performance. The regularization parameter was selected using cross-validation, with the λ_{lse} model chosen to balance model simplicity and accuracy. Figure @q1-lasso-cv-plot presents the cross-validation curve used to select the optimal regularization parameter. While λ_{min} achieves the lowest prediction error, the λ_{lse} model provides comparable performance with substantially fewer predictors and is therefore selected for interpretation. The resulting set of non-zero coefficients identifies the most influential variables in predicting income.

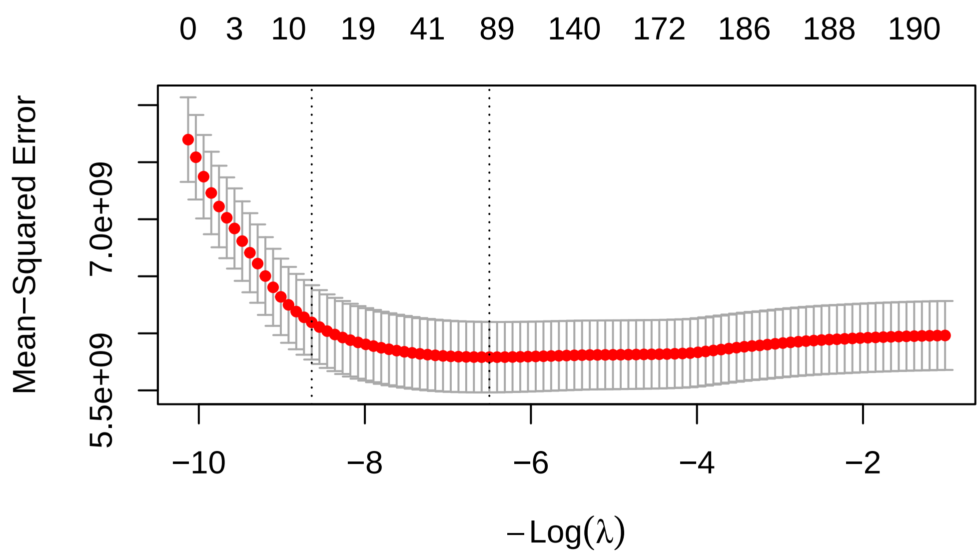


Figure 4: Cross-validation curve for the LASSO model showing mean squared error across values of the regularization parameter lambda.

Q1: Inferences

In the multiple linear regression model, several predictors show statistically significant associations with income. Education has a strong positive effect, indicating that higher educational attainment is associated with increased income, holding other variables constant. Age exhibits a nonlinear relationship with income, with the quadratic term confirming that earnings

Variable	Coefficient
AGEP	133.39
SEXFemale	-10,114.75
MARNever married	-5,842.34
RELSHIPP	-773.88
SCHL	2,776.24
COWSelf-emp not incorporated	3,803.42
WKHP	1,270.42
WKWN	520.95
PUMA05114	26,557.74
PUMA06721	755.32
TENRented	-11,765.93
HISP_binaryNot Hispanic	11,661.62

increase early in life and level off or decline at older ages. Gender and race also show statistically significant differences, suggesting persistent disparities in income across demographic groups. While the overall explanatory power of the model is modest, these results indicate that key demographic and socioeconomic factors contribute meaningfully to variation in income. The LASSO results support these findings by selecting a similar subset of variables, reinforcing the importance of education, labor supply, and demographic characteristics as key predictors of income.

Question2:

Q2:

Q2:

Question3:

Q3:

Q3:

Obstacles

Conclusions

Our analysis shows that income can be partially predicted using age, education, gender, and race. Education is the strongest predictor, with higher levels of education associated with substantially higher income. Age has a nonlinear relationship with income, with earnings increasing early in life, peaking in mid-career, and declining slightly thereafter. Gender and race also show statistically significant associations with income, even after controlling for other variables.

Despite these significant relationships, the overall explanatory power of the model is modest ($R^2 = 0.12$), indicating that we may need to consider other variables as determinants of income. Going forward there are additional factors to consider. One would be exploring additional variables for positive associations with income that could be added to the model. Another option is testing interaction terms. We could also consider whether gender and race are impactful enough that we should really be developing separate models in order to accurately capture the full impact these categorical variables have on all other aspects.

Citations

1. U.S. Census Bureau. (2026). 2020–2024 American Community Survey 5-year Public Use Microdata Sample (PUMS). U.S. Department of Commerce. https://www.census.gov/programs-surveys/acs/microdata.html

2. Mason, H., & Wiggins, C. (2010). A Taxonomy of Data Science. (Archived manuscript / teaching document). Accessed on 4/10/2026 https://introdatasci.dlilab.com/pdf/A_Taxonomy_of_Data_Science.pdf

Appendix

```
knitr::opts_chunk$set(
  warning = FALSE,
  message = FALSE)
#install.packages("ggfortify")


library(tidyverse)
library(patchwork)
library(ggplot2)
library(ggfortify)
library(glmnet)
library(faraway)
library(car)
library(gt)
# reticulate::virtualenv_create("r-reticulate")
# reticulate::virtualenv_install("r-reticulate", c("requests", "pandas"))
```

```
# file.edit("~/.Rprofile")
# add in this
# Sys.setenv(RETICULATE_PYTHON_ENV = "r-reticulate")
# run the following restart the python session if needed
# reticulate::py_run_string("import gc; gc.collect()")
import requests
import json
from pathlib import Path

METADATA_PATH = Path("data/pums_variables_metadata.json")

variables = [ # added POBP, JWTRNS, NRC
    "WAGP", "PINCP", "PERNP", "POVPIP", "HICOV", "ESR",
    "AGEP", "SEX", "RAC1P", "HISP", "CIT", "YOEP", "MAR", "RELSHIPP",
    "POBP", "SCHL", "FOD1P", "SCH", "COW", "WKHP", "WKWN", "NAICSP", "OCCP",
    "WRK", "JWMNP", "JWTRNS", "PUMA", "MIG", "MIGSP", "POWSP",
    "DIS", "DPHY", "DREM", "DEAR", "DEYE","HINCP", "HHT", "TEN", "VEH",
    "NRC", "RETP", "SSP", "SSIP", "PAP", "SEMP",
    "PWGTP", "ADJINC",
]

BASE_URL = "https://api.census.gov/data/2024/acs/acs1/pums/variables/{}.json"

if METADATA_PATH.exists():
    print(f"Loading metadata from cache: {METADATA_PATH}")
    with open(METADATA_PATH) as f:
        metadata = json.load(f)
else:
    print("Fetching variable metadata from Census API...")
    METADATA_PATH.parent.mkdir(parents=True, exist_ok=True)
    metadata = {}

    for var in variables:
        response = requests.get(BASE_URL.format(var), timeout=30)
        if response.status_code == 200:
            metadata[var] = response.json()
            print(f"  OK: {var}")
        else:
            print(f"  FAILED: {var} - {response.status_code}")

    with open(METADATA_PATH, "w") as f:
        json.dump(metadata, f, indent=2)
```

```
    print(f"\nSaved metadata for {len(metadata)} variables to {METADATA_PATH}")

# Preview one variable
print(json.dumps(metadata.get("WAGP", {}), indent=2))
# --- Scrub Start ---

import requests
import pandas as pd
from pathlib import Path

API_KEY = "a8f34453c39e419470f405353250f662a508d821"
YEAR = 2024
STATE = "41"
CACHE_PATH = Path("data/pums_oregon_2024.csv")

variables = [
    # Outcomes
    "WAGP", "PINCP", "PERNP", "POVPIP", "HICOV", "ESR",
    # Core demographics
    "AGEP", "SEX", "RAC1P", "HISP", "CIT", "YOEP", "MAR", "RELSHIPP",
    "POBP",                                                         # added POBP
    # Education
    "SCHL", "FOD1P", "SCH",
    # Labor market
    "COW", "WKHP", "WKWN", "NAICSP", "OCCP", "WRK", "JWMNP", "JWTRNS", # added
    # Geography
    "PUMA", "MIG", "MIGSP", "POWSP",
    # Disability & health
    "DIS", "DPHY", "DREM", "DEAR", "DEYE",
    # Household context
    "HINCP", "HHT", "TEN", "VEH", "NRC",                              # added NRC
    # Income components
    "RETP", "SSP", "SSIP", "PAP", "SEMP",
    # Weights & adjustments
    "PWGTP", "ADJINC","NP",
]

if CACHE_PATH.exists():
    print(f"Loading from local cache: {CACHE_PATH}")
    pums = pd.read_csv(CACHE_PATH, low_memory=False)
    print(f"Loaded: {len(pums):,} rows, {pums.shape[1]} columns")
else:
```

```
    print(f"Cache not found - fetching from Census API...")
    CACHE_PATH.parent.mkdir(parents=True, exist_ok=True)
    url = (
        f"https://api.census.gov/data/{YEAR}/acs/acs5/pums"
        f"?get={','.join(variables)}"
        f"&for=state:{STATE}"
        f"&key={API_KEY}"
    )
    response = requests.get(url, timeout=60)
    print(f"Status code: {response.status_code}")
    if response.status_code == 200:
        data = response.json()
        pums = pd.DataFrame(data[1:], columns=data[0])
        pums.to_csv(CACHE_PATH, index=False)
        print(f"Saved to {CACHE_PATH}: {len(pums):,} rows, {pums.shape[1]} columns")
    else:
        print(f"Error: {response.text[:500]}")
pums <- read_csv("data/pums_oregon_2024.csv", show_col_types = FALSE)

#glimpse(pums)
pums_clean <- pums |>

  # --- Remove group quarters records ---
  filter(HINCP != -60000) |>

  # --- Apply income adjustment factor ---
  mutate(
    across(c(WAGP, PINCP, PERNP, HINCP, RETP, SSP, SSIP, PAP, SEMP),
           ~ .x * ADJINC)
  ) |>

  # --- Recode all N/A sentinels to NA ---
  mutate(
    # Income vars: age-based N/A
    WAGP   = na_if(WAGP,  -1 * ADJINC),
    PERNP  = na_if(PERNP, -10001 * ADJINC),
    PINCP  = na_if(PINCP, -19999 * ADJINC),
    RETP   = na_if(RETP,  -1 * ADJINC),
    SSP    = na_if(SSP,   -1 * ADJINC),
    SSIP   = na_if(SSIP,  -1 * ADJINC),
    PAP    = na_if(PAP,   -1 * ADJINC),
    SEMP   = na_if(SEMP,  -10001 * ADJINC),
```

```r
    # Labor market: 0 = N/A (under 16 or did not work)
    WKHP   = na_if(WKHP, 0),
    WKWN   = na_if(WKWN, 0),
    JWMNP  = na_if(JWMNP, 0),
    # Other sentinels
    POVPIP = na_if(POVPIP, -1),
    ESR    = na_if(ESR, 0),
    YOEP   = na_if(YOEP, 1937),
    HHT    = na_if(HHT, 0),
    TEN    = na_if(TEN, 0),
    VEH    = na_if(VEH, -1),
    NRC    = na_if(NRC, -1),
    DPHY   = na_if(DPHY, 0),
    DREM   = na_if(DREM, 0),
    NP     = na_if(NP, 0),
    # Labor class and work status: 0 = N/A
    COW    = na_if(COW, 0),
    WRK    = na_if(WRK, 0),
  ) |>

  # --- Recode binary vars to logical (1=Yes, 2=No) ---
  mutate(
    across(c(DPHY, DREM, DEAR, DEYE, DIS), ~ .x == 1),
    HICOV = HICOV == 1,
  ) |>

  # --- Recode categorical vars as factors with labels from metadata ---
  mutate(
    SEX = factor(SEX, levels = c(1, 2),
                 labels = c("Male", "Female")),

    RAC1P = factor(RAC1P, levels = 1:9,
                   labels = c("White alone", "Black or African American alone",
                              "American Indian alone", "Alaska Native alone",
                              "AIAN tribes specified", "Asian alone",
                              "Native Hawaiian and Other Pacific Islander alone",
                              "Some other race alone", "Two or More Races")),

    HISP_binary = factor(if_else(HISP == "01", "Not Hispanic", "Hispanic")),

    CIT = factor(CIT, levels = 1:5,
                 labels = c("Born in US", "Born in PR/territories",
```

```
                          "Born abroad of US citizen parent",
                          "Naturalized citizen", "Not a US citizen")),

    MAR = factor(MAR, levels = 1:5,
                 labels = c("Married", "Widowed", "Divorced",
                            "Separated", "Never married")),

    ESR = factor(ESR, levels = 1:6,
                 labels = c("Civilian employed at work",
                            "Civilian employed not at work",
                            "Unemployed",
                            "Armed Forces at work",
                            "Armed Forces not at work",
                            "Not in labor force")),

    SCH = factor(SCH, levels = 1:3,
                 labels = c("Not enrolled", "Public school", "Private school")),

    COW = factor(COW, levels = 1:9,
                 labels = c("Private for-profit", "Private nonprofit",
                            "Local govt", "State govt", "Federal govt",
                            "Self-emp incorporated", "Self-emp not incorporated",
                            "Working without pay", "Unemployed last 5 years")),

    MIG = factor(MIG, levels = 1:3,
                 labels = c("Same house", "Outside US", "Different house in US")),

    HHT = factor(HHT, levels = 1:7,
                 labels = c("Married couple household",
                            "Other family: male householder no spouse",
                            "Other family: female householder no spouse",
                            "Nonfamily: male householder living alone",
                            "Nonfamily: male householder not living alone",
                            "Nonfamily: female householder living alone",
                            "Nonfamily: female householder not living alone")),

    TEN = factor(TEN, levels = 1:4,
                 labels = c("Owned with mortgage", "Owned free and clear",
                            "Rented", "Occupied without payment")),

    WRK = factor(WRK, levels = 1:2,
                 labels = c("Worked last week", "Did not work last week")),
```

```r
    JWTRNS = factor(JWTRNS, levels = 0:12,
                    labels = c("N/A", "Car truck or van", "Bus",
                               "Subway or elevated rail",
                               "Long-distance train or commuter rail",
                               "Light rail streetcar or trolley",
                               "Ferryboat", "Taxi or ride-hailing",
                               "Motorcycle", "Bicycle", "Walked",
                               "Worked from home", "Other method")),

    POBP_binary = factor(case_when(
      as.numeric(POBP) <= 56  ~ "Born in US",
      as.numeric(POBP) == 72  ~ "Born in PR/territories",
      TRUE                    ~ "Foreign born"
    )),
  ) |>

  # --- Drop redundant/replaced columns ---
  select(-ADJINC, -HISP, -POBP, -state) |>

  # --- Ensure valid person weight ---
  filter(!is.na(PWGTP), PWGTP > 0)

#glimpse(pums_clean)
# --- Note ---
# after data pull and scrub in Python from the API
# the following was a proposed EDA method
# the group did chose to work in a different direction for EDA
# we have left if here for possible future use

library(tidyverse)
library(skimr)

# 1. overall summary
#skim(pums_clean)
# Wages among earners only

# Log wages

# Total personal income
# Q1/Q3: wage statistics by key groups

# Q2: health insurance coverage rate
```

```
# Q2: uninsured rate by citizenship
# --- End EDA proposal ---

library(corrr)
p1 <-ggplot(data = pums_clean, aes(x = PINCP)) +
  geom_histogram(binwidth = 30000) +
  labs(
    y = "Count",
    x = "Personal Income"
  )


p2 <-ggplot(data = pums_clean, aes(x = log(pums_clean$PINCP)+1)) +
  geom_histogram(binwidth = 1) +
  labs(
    y = "Count",
    x = "log() of Personal Income"
  )

p1 + p2 + plot_layout(ncol = 2)
# This takes a long time to run, adding if statement
# so it only gets run when needed.
outfile <- "./eda-incomevage-plot.png"

if (!file.exists(outfile)) {
  incomevage_plot <- ggplot(pums_clean, aes(x = AGEP, y = PINCP)) +
    geom_point(alpha = 0.2) +
    geom_smooth(method = "loess") +
    scale_y_continuous(labels = label_number()) +
    labs(title = "Census Data: Personal Income vs Age")
    ggsave(outfile, plot = incomevage_plot, width = 10, height = 6, dpi = 300)
}
# This takes a long time to run, adding if statement
# so it only gets run when needed.
outfile <- "./eda-incomeved-plot.png"

if (!file.exists(outfile)) {
  incomeved_plot <- ggplot(pums_clean, aes(x = SCHL, y = PINCP)) +
    geom_point(alpha = 0.2) +
    geom_smooth(method = "loess") +
    scale_y_continuous(labels = label_number()) +
    labs(title = "Income vs Education", x = "Years of School (Education)", y = "Income")
```

```r
  ggsave(outfile, plot = incomeved_plot, width = 10, height = 6, dpi = 300)
}
ggplot(pums_clean, aes(x = factor(SEX), y = PINCP)) +
  geom_boxplot() +
  coord_cartesian(ylim = c(0, 200000)) +
  scale_x_discrete(labels = c("Male", "Female")) +
  labs(
    title = "Income by Gender",
    x = "Gender",
    y = "Income"
  ) +
  theme_minimal()
ggsave("./eda-incomevgender-plot.png", width = 10, height = 6, dpi = 300)
ggplot(pums_clean, aes(x = factor(RAC1P), y = PINCP)) +
  geom_boxplot() +
  coord_cartesian(ylim = c(0, 200000)) +
  labs(
    title = "Income by Race",
    x = "Race",
    y = "Income"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_x_discrete(labels = c(
    "White alone" = "White",
    "Black or African American alone" = "Black",
    "American Indian alone" = "AIAN",
    "Alaska Native alone" = "Alaska Nat.",
    "AIAN w/ tribes specified" = "AIAN (tribe)",
    "Asian alone" = "Asian",
    "Native Hawaiian and Other Pacific Islander alone" = "NHPI",
    "Some other race alone" = "Other",
    "Two or More Races" = "Two or More Races"
  ))
ggsave("./eda-incomevrace-plot.png", width = 10, height = 6, dpi = 300)
# Urban/Rural is hard to codify in the data, so it may be best to not include it.


model1 <- lm(PINCP~AGEP+I(AGEP^2)+SCHL+factor(SEX)+factor(RAC1P)+factor(POBP_binary) + NP, da
summary(model1)

model2 <- lm(PINCP~AGEP+I(AGEP^2)+SCHL+factor(SEX)+factor(RAC1P), data=pums_clean)
```

```
summary(model2)

vif(model1)

model3 <- lm(PINCP~AGEP+SCHL+factor(SEX)+factor(RAC1P), data=pums_clean)
summary(model2)

anova(model2, model3)


vif(model2)
# Urban/Rural is hard to codify in the data, so it may be best to not include it.

model1_log <- lm(log(PINCP + 1)~AGEP+I(AGEP^2)+SCHL+factor(SEX)+factor(RAC1P)+factor(POBP_bin
summary(model1_log)

model2_log <- lm(log(PINCP + 1)~AGEP+I(AGEP^2)+SCHL+factor(SEX)+factor(RAC1P), data=pums_clea
summary(model2_log)

vif(model1_log)

model3_log <- lm(log(PINCP + 1)~AGEP+SCHL+factor(SEX)+factor(RAC1P), data=pums_clean)
summary(model3_log)

anova(model2_log, model3_log)

vif(model2_log)


# Use all variables except PINCP(response) The other income related variables
form <- PINCP ~ . -OCCP -FOD1P - NAICSP +I(AGEP^2) -PERNP -WAGP -PINCP -PERNP -POVPIP -HICOV

# Keep only rows with complete data for variables in the model
pums_clean_no_na <- model.frame(form, data = pums_clean, na.action = na.omit)

# Create predictor matrix for glmnet
# The -1 removes the intercept because glmnet adds its own intercept
#X <- model.matrix(
#  PINCP ~ AGEP + I(AGEP^2) + SCHL + factor(SEX) +
#    factor(RAC1P) + factor(POBP_binary) + NP,
#  data = pums_clean_no_na
#)[, -1]
```

```r
X <- model.matrix(form, data = pums_clean_no_na)[, -1]

# Create response vector
y <- as.numeric(pums_clean_no_na$PINCP)

##### Lasso Estimates
lasso <- glmnet(X, y)
plot(lasso)

lasso.cv <- cv.glmnet(X, y)
lasso1 <- glmnet(X, y, lambda = lasso.cv$lambda.min)
lasso1$beta
lasso2 <- glmnet(X, y, lambda = lasso.cv$lambda.1se)
lasso2$beta

#coef(lasso.cv)
# non-log model
plot(model2, which = 1)
#plot(model2, which = 2)

#model_log
plot(model2_log, which = 1)
#plot(model2_log, which = 2)
# Original model
mp1 <- autoplot(model2, which = 2) + ggtitle("QQ Plot: Original Income")

# Log Model
mp2 <- autoplot(model2_log, which = 2) + ggtitle("QQ Plot: Log Income")

combined_plot <- mp1 + mp2 + plot_layout(ncol = 2)
ggsave("q1-model-plots2.png", plot = combined_plot,
       width = 10, height = 6, dpi = 300)
# lambda.min
coef_min <- coef(lasso.cv, s = "lambda.min")

# lambda.1se
coef_1se <- coef(lasso.cv, s = "lambda.1se")

# remove intercept + keep nonzero
vars_min <- rownames(coef_min)[coef_min[,1] != 0][-1]
vars_1se <- rownames(coef_1se)[coef_1se[,1] != 0][-1]
```

```r
# Show both models in full
#vars_min
#vars_1se

# Quick compare of size of the models
#length(vars_min)
#length(vars_1se)
plot(lasso.cv)
# Get coefficients at lambda.1se
coef_1se <- coef(lasso.cv, s = "lambda.1se")

# Convert to regular vector with names
coef_vec <- as.vector(coef_1se)
names(coef_vec) <- rownames(coef_1se)

# Remove intercept and keep only non-zero coefficients
selected <- coef_vec[-1][coef_vec[-1] != 0]
#selected


selected_df <- data.frame(
  variable = names(selected),
  coefficient = selected
)

selected_df %>%
  gt() %>%
  fmt_number(columns = coefficient, decimals = 2) %>%
  cols_label(
    variable = "Variable",
    coefficient = "Coefficient"
  )
```